

Place Value, Rounding and Estimating — Paper B

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to be handled on paper.
- The decimal point does not move. The **digits** move: one column left for each $\times 10$, one column right for each $\div 10$.
- **$\times 0.1$ is the same as $\div 10$** , and **$\div 0.1$ is the same as $\times 10$** . Rewrite the question that way before you calculate.
- Read every rounding instruction twice: **decimal places** and **significant figures** give different answers on the same number.
- Write a zero before the decimal point (0.47, never .47), and keep placeholder zeros (38 000, never 38).
- The mark for each question is shown in square brackets on the right.

SECTION 1 – MULTIPLYING AND DIVIDING BY 0.1

- 1.** Complete the table. Each empty box is worth one mark. One of these columns makes the numbers smaller and the other makes them bigger — decide which before you start. [6]

Number	$\times 0.1$	$\div 0.1$
4.7		
82		
0.5		

Rewrite each one first: $\times 0.1$ is $\div 10$, and $\div 0.1$ is $\times 10$.

- 2.** Work out: [1]
- (a)** 6.9×0.01

- (b)** $0.4 \div 0.01$ [1]

- (c)** 250×0.1 [1]

(d) $7.3 \div 0.1$

[1]

3. Fill in the missing number in each.

(a) $8.4 \times \underline{\hspace{1cm}} = 0.084$

[1]

(b) $5 \div \underline{\hspace{1cm}} = 500$

[1]

SECTION 2 – DECIMAL PLACES AND SIGNIFICANT FIGURES

4. Round **0.083529** to:

(a) 2 decimal places

[1]

(b) 3 decimal places

[1]

(c) 2 significant figures

[1]

(d) 3 significant figures

[1]

5. Round **74 851** to:

(a) the nearest 100

[1]

(b) the nearest 1000 [1]

(c) 2 significant figures [1]

(d) 3 significant figures [1]

6. Write down **any** number that rounds to:

(a) 60, to the nearest 10 [1]

(b) 4.7, to 1 decimal place [1]

(c) 0.03, to 2 decimal places [1]

(d) 200, to 1 significant figure [1]

SECTION 3 – ESTIMATING

7. A customer buys 4 shirts at 18.90 rupees each, 3 notebooks at 6.20 rupees each, and 2 bags at 44.50 rupees each.

(a) Estimate the total bill by rounding each price to 1 significant figure. [2]

- (b)** The till prints a total of 1782.20 rupees. Explain how your estimate shows this must be wrong. [1]

8. This question is about 46×78 .

- (a)** Estimate the answer by rounding each number to 1 significant figure. [1]

- (b)** Work out the exact answer. [1]

- (c)** State whether your estimate is bigger or smaller than the exact answer, and explain why. [1]

End of paper. Check every answer has a unit or a form the question actually asked for.